EXAM SKILLS WORKSHEET

Routing Table Lab

Interpreting a routing table to work out how a router forwards traffic. Covers what each field means, the rules a router applies, and exam-style practice. (Spec: purpose of routing-table data; interpret data to forward traffic.)

What a routing table stores

A routing table stores whole **networks**, not single addresses. Read each entry left to right:

```
S   192.168.1.128/26   [1/0]   via 10.0.0.6,   Gi0/1
```

Field	Meaning
S — code letter	How the route was added (C connected, S static, O OSPF, R RIP, D EIGRP). Not how good it is.
192.168.1.128 — network ID	The network this entry represents.
/26 — subnet mask	How many bits are the network part (255.255.255.192).
via 10.0.0.6 — next hop	The IP of the next router to forward to. A directly connected (C) route has **no next hop**.
Gi0/1 — interface	The router's own port the packet leaves by.

How a router chooses — in order

1. **Longest prefix match.** If more than one entry matches, use the one with the longest (most specific) mask — a /26 beats a /24 for the same address.
2. **Administrative distance.** If the same network is learned from two sources, trust the lower AD (static = 1 beats OSPF = 110).
3. **Default route (0.0.0.0/0).** Matches everything, but wins only when nothing more specific does — the way out to the internet.
4. **No match and no default** → the router **drops** the packet.

Static vs dynamic

Static: added by hand (`ip route …`), never changes unless edited — simple, secure, predictable; good for small networks. **Dynamic:** built automatically by a protocol (RIP/OSPF/EIGRP), updates itself when links change — better for large, growing networks, but needs more processing and care.

Worked example

Table:

```
C   10.0.0.0/30        directly connected, Gi0/0
S   192.168.1.0/24     via 10.0.0.2,   Gi0/0
S   192.168.1.128/26   via 10.0.0.6,   Gi0/1
O   172.16.0.0/16      via 10.0.0.2,   Gi0/0
S*  0.0.0.0/0          via 203.0.113.1, Gi0/2
```

Packet to 192.168.1.130: both the /24 and the /26 match. Longest prefix match picks the **/26** → forward **via 10.0.0.6, out Gi0/1**.

Packet to 8.8.8.8: no specific entry matches → use the **default route 0.0.0.0/0** → out Gi0/2.

Exam tip: always line the destination up against each network's mask first, then apply longest prefix match. Write the winning entry, then its next hop and interface.

Using the online lab

- **Read the fields** — the anatomy of an entry, the code letters, and the decision rules.
- **Worked example** — step through the matching logic; switch the destination to see each case (specific match, longest prefix, default route, directly connected).
- **Practice** — ten exam-style questions with the reasoning shown for every answer; best score saved.